Hamiltonians

$$i\hbar \frac{\partial(\psi(r,t))}{\partial t} = \hat{H}\psi(r,t)$$

$$\hat{H} = \frac{\hat{p}^2}{2m} + \hat{V}(r) = \frac{\hbar^2}{2m}\nabla^2 + \hat{V}(r)$$

$$\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$$

$\hat{H} = \hat{H}(\hat{x},\hat{p})$ ⇝ Time independent Hamiltonian

$$\psi(r,t) = e^{-iHt/\hbar} \cdot \psi(r,0)$$

For $\hat{H}\psi(r,0) = E\psi(r,0)$

E → Energy → Eigenvalue → Real

Hermitian → $\hat{H} = \hat{H}^\dagger$

$$U(t) = e^{-iHt/\hbar} \rightarrow \hat{H} = \hat{H}^\dagger \rightarrow U^\dagger U = I$$

$$|\psi(t)\rangle = U(t)\,|\psi(0)\rangle$$

→ Spectral Decomposition

$$\hat{H} = \sum_n E_n |E_n\rangle\langle E_n| \rightarrow |\psi(0)\rangle = \sum_n c_n |E_n\rangle$$

$$c_n = \langle E_n | \psi(0)\rangle$$

$$|\psi(t)\rangle = e^{-iHt/\hbar} \sum_n c_n |\epsilon_n\rangle$$

$$e^{-i\hat{H}t/\hbar} |\epsilon_n\rangle = e^{-i\epsilon_n t/\hbar} |\epsilon_n\rangle$$

$$|\psi(t)\rangle = \sum_n c_n e^{-iE_n t/\hbar} |\epsilon_n\rangle$$

$$|\psi(0)\rangle = \frac{|\epsilon_1\rangle + |\epsilon_2\rangle}{\sqrt{2}} \rightarrow |\psi(t)\rangle = \frac{e^{-iE_1 t/\hbar}|\epsilon_1\rangle + e^{-i\epsilon_2 t/\hbar}|\epsilon_2\rangle}{\sqrt{2}}$$

Relative Phase $\rightarrow e^{-i\Delta E t/\hbar}$ $\rightarrow \Delta\epsilon = \epsilon_1 - \epsilon_2$

$$\hat{H} = \hat{H} + \hat{C} \qquad \hat{C} \rightarrow \text{constant}$$

$$U(t) = \sum_n e^{-iE_n t/\hbar} |E_n\rangle\langle E_n|$$

→ Quantum Gates as Hamiltonians

→ X Gate

$$\hat{H} = \frac{\hbar\omega}{2} X \qquad U(t) = e^{-i\omega X t/2} = \cos\left(\frac{\omega t}{2}\right) \cdot I - i \sin\left(\frac{\omega t}{2}\right) X$$

At $t = \frac{\pi}{\omega}$ → $U(t) = -iX \equiv X$

↳ we evolve system

Y Gate

$$\hat{H} = \frac{\hbar\omega}{2} Y$$

2-Gate

$$\hat{H} = \frac{\hbar\omega}{2} 2$$

→ Hamiltonian for e^- in Magnetic Field

$$U = -\mu \cdot B \qquad \mu = \gamma S \qquad U = -\gamma S \cdot B$$

$$S_i = \frac{\hbar}{2}\sigma_i \qquad B = (B_x, 0, 0)$$

$$\hat{H} = -\frac{\hbar \gamma B_x}{2} \cdot X \quad \rightarrow \omega = \gamma \cdot B_x$$

In general → $\hat{H} = \frac{\hbar\Omega}{2}(\vec{n}\cdot\vec{\sigma})$

$$\Omega = \gamma\sqrt{B_x^2 + B_y^2 + B_z^2}$$

$$\vec{n} = \frac{\vec{B}}{|\vec{B}|}$$

Time Independent Hamiltonian

Total Evolution → $e^{-iH(t_3)\cdot\Delta t/\hbar} \cdot e^{-iH(t_2)\cdot\Delta t/\hbar} \cdot e^{-iH(t_1)\cdot\Delta t/\hbar}$ ——

$$= U(t) = \mathcal{T} e^{-i\hat{H}' t/\hbar}$$

$$\hat{H}' = \frac{\int_0^t H(t)\cdot dt}{t}$$

$\mathcal{T}$ → Time ordering operator

Needed as → $\left(H(t_i), H(t_j)\right) \neq 0$

Dyson Series

$$U(t) = I + \left(\frac{-i}{\hbar}\right)\int_0^t H(t_1)\cdot dt_1 + \left(\frac{-i}{\hbar}\right)^2 \int_0^t dt_1 \cdot \int_0^t H(t_1)\cdot H(t_2)\cdot dt_2$$

$$U(t,t_0) = \sum_{n=0}^{\infty} \frac{1}{n!}\left(\frac{-i}{\hbar}\right)^n \int_{t_0}^{t} dt_1 \cdots \int_{t_0}^{t} dt_n \; T\left[H(t_n)\cdots H(t_1)\right]$$

Von Neumann Equation

$$i\hbar \frac{\partial \rho}{\partial t} = \left[\hat{H}, \rho\right]$$

$$\hat{H}|\epsilon\rangle = \epsilon|\epsilon\rangle \rightarrow \left[\hat{H}, \rho\right] = 0 \rightarrow \frac{\partial \rho}{\partial t} = 0$$

$$\rho(t) = e^{-i\epsilon t/\hbar}|\epsilon\rangle\langle\epsilon| e^{i\epsilon_n t/\hbar}$$

$$= |\epsilon\rangle\langle\epsilon| = \rho(0)$$

Considering Environment

$$\rho_s = Tr_\epsilon(\rho)$$